Supplementary Materials for

Neural dynamics of updating social impressions during movie watching

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Supplementary Figures

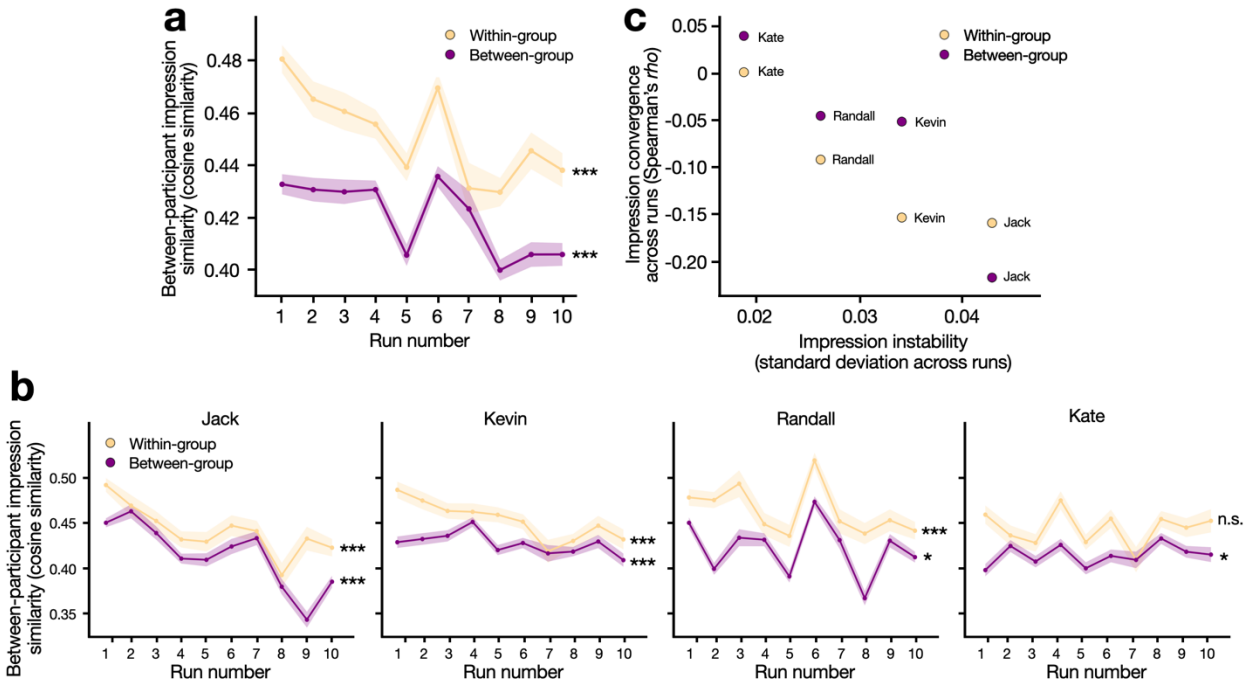


Figure S1. Social impressions diverge across individuals over time. (a) How between-participant impression similarity (cosine similarity between the USE embeddings of transcriptions of character descriptions, averaged across the four main characters) changes as a function of run number. Yellow indicates the mean impression similarity across participant pairs that watched the events in the same scrambled order (within-group), whereas purple indicates results from participant pairs that watched in different scrambled orders (between-group). **(b)** How between-participant impression similarity changes as a function of run number separately for each of the four characters. **(c)** Impression instability tracks between-participant impression convergence both within- and between-groups. Impression instability is measured as the standard deviation of between-run impression updates (1 - cosine similarity) across the 9 run transitions. Impression convergence is measured as the Spearman correlation between impression similarity and the run number. n.s., not significant; *, $p < 0.05$; ***, $p < 0.001$

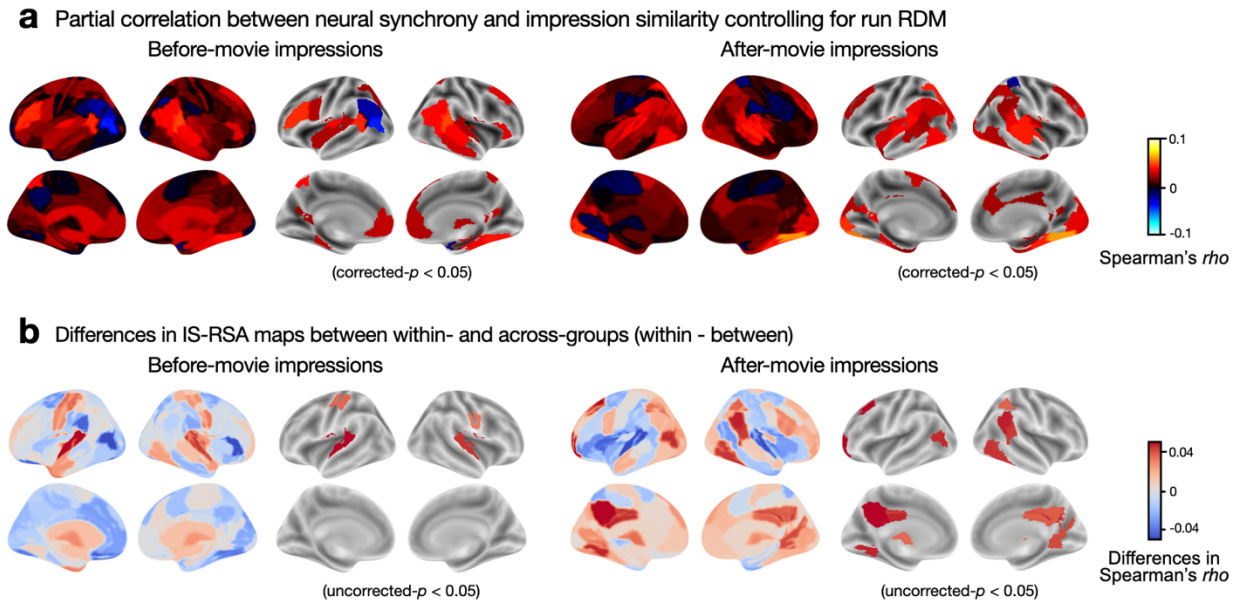


Figure S2. **Shared narrative history modulates the relationship between neural synchrony and impression similarity. (a)** Partial correlations between neural synchrony and impression similarity, controlling for the representational dissimilarity matrix (RDM) indicating whether participant pairs belonged to the same randomized sequence group. The analysis was conducted separately for pre-movie impressions (left) and post-movie impressions (right). **(b)** Difference between IS-RSA effects estimated for within-group and between-group participant pairs. A two-sided test assessed whether the strength of the neural–impression similarity relationship differed depending on whether participants shared the same randomized narrative sequence. Red indicates stronger effects within groups ($r_{within} > r_{between}$), whereas blue indicates the reverse.

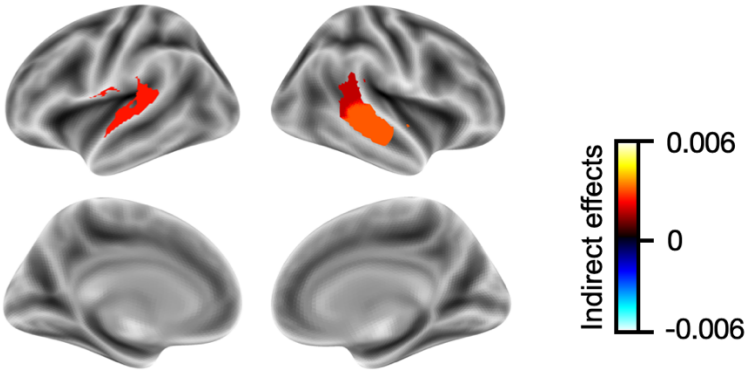


Figure S3. Neural synchrony mediates the relationship between prior and subsequent impressions. Regions whose neural synchrony during movie-watching exhibit significant (uncorrected-$p < 0.05$) or marginal (uncorrected-$p < 0.10$) mediation effect on the relationships between before- and after-movie impressions, before correcting for multiple comparisons. This is a thresholded version of Fig. 3b.

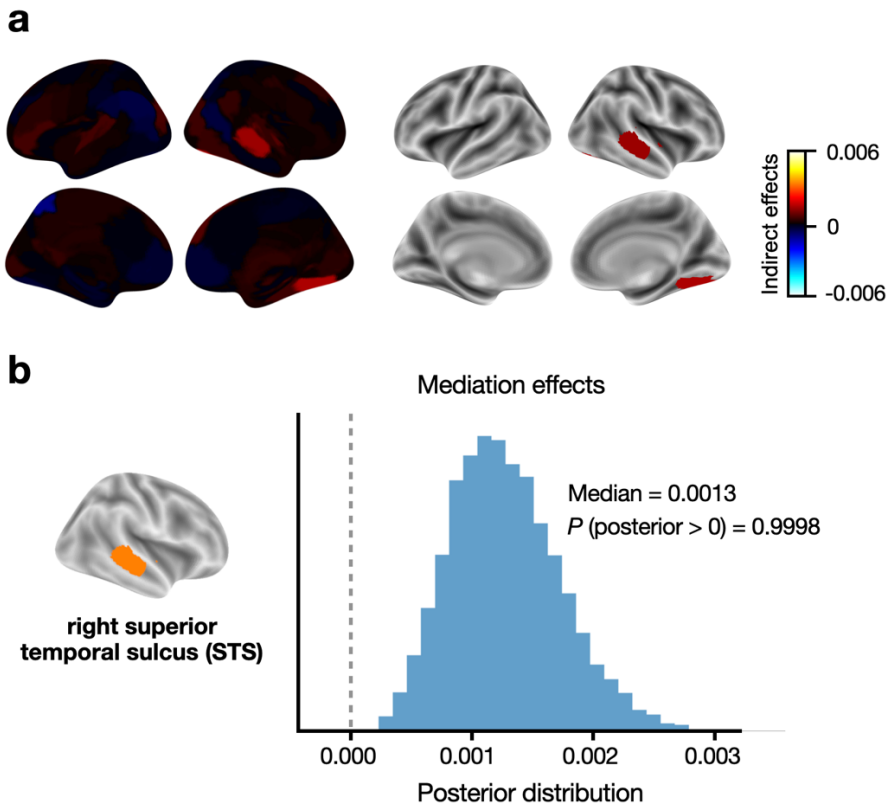


Figure S4. Neural synchrony in the right STS mediates impression updating after controlling for narrative order. (a) Mediation analysis testing whether neural synchrony mediated impression updating after controlling for run RDM. The analysis followed the original pipeline, with the run RDM included as an additional covariate. **(b)** The posterior distribution of the mediation effects in the right STS.

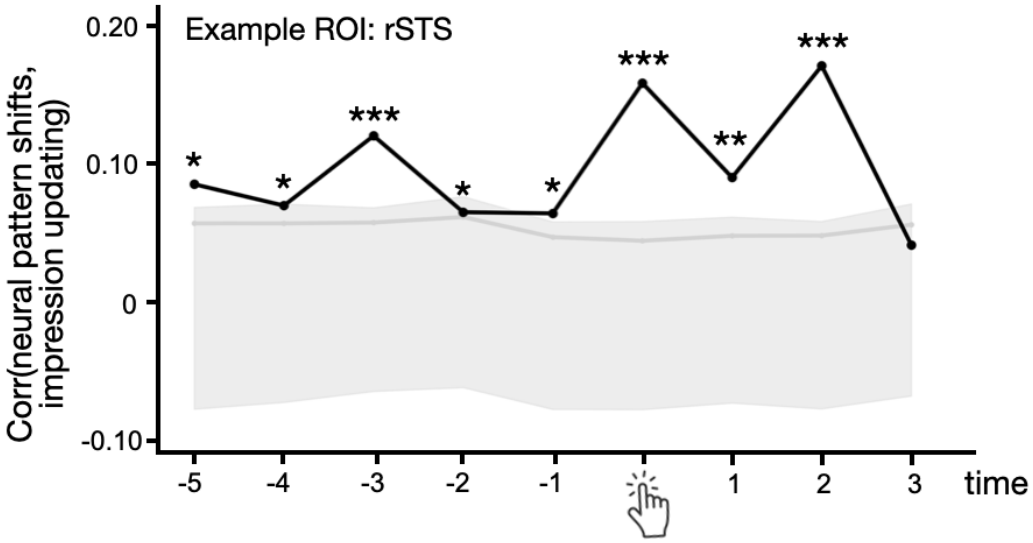


Figure S5. Neural pattern shifts in the right STS track impression updating near moments of character insight. Correlations between neural pattern shifts in the right superior temporal sulcus and impression updating as a function of time relative to aha button presses. The gray area represents a null distribution of these correlations from non-insight moments. The gray line indicates the top 5% *r*-value in the null distribution. The asterisks indicate statistical significance comparing the *r*-values to the null distribution with one-tailed tests at each TR (before correction). *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$.

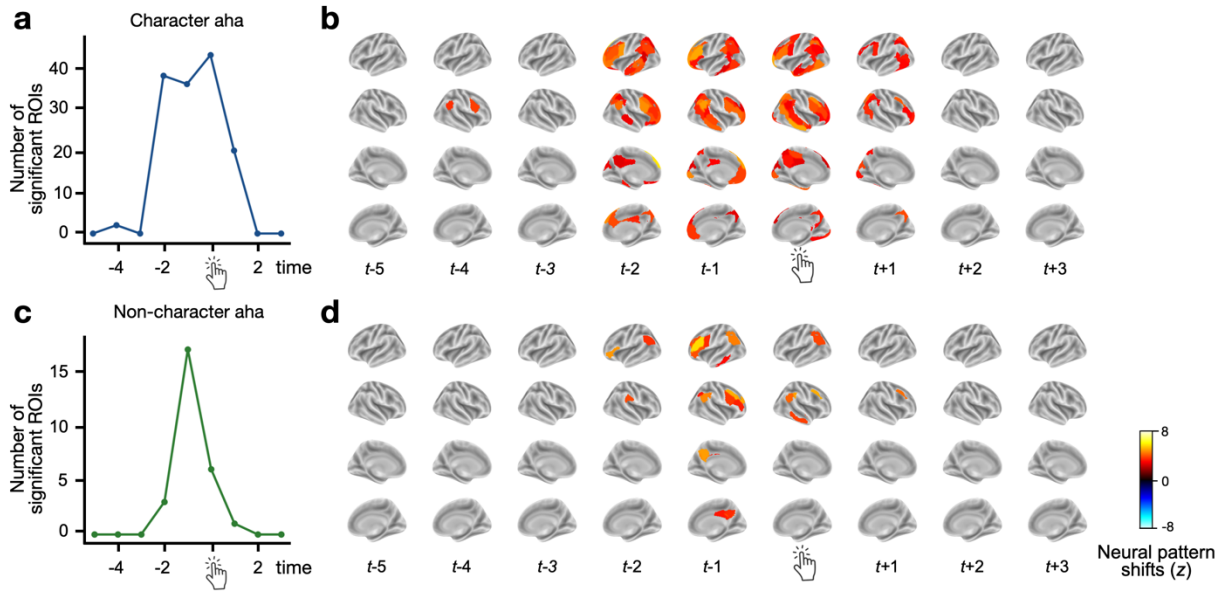


Figure S6. Neural pattern shifts occur near character and non-character insight moments. **(a)** Number of ROIs with significant neural pattern shifts near *character aha* moments as a function of time. **(b)** Whole-brain neural pattern shifts as a function of time relative to *character aha* button presses. Brain maps plot the *z*-statistics comparing neural pattern shifts near character aha moments with those during non-aha moments. Regions with significant neural pattern shifts were plotted, corrected for multiple comparisons across 116 ROIs and 9 TRs (corrected-$p < 0.05$). This figure is identical to Fig. 4d. **(c)** Number of ROIs with significant neural pattern shifts near *non-character aha* moments as a function of time. **(d)** Whole-brain neural pattern shifts as a function of time relative to *non-character aha* button presses. Brain maps plot the *z*-statistics comparing neural pattern shifts near non-character aha moments with those during non-aha moments. Regions with significant neural pattern shifts were plotted, corrected for multiple comparisons across 116 ROIs and 9 TRs (corrected-$p < 0.05$).

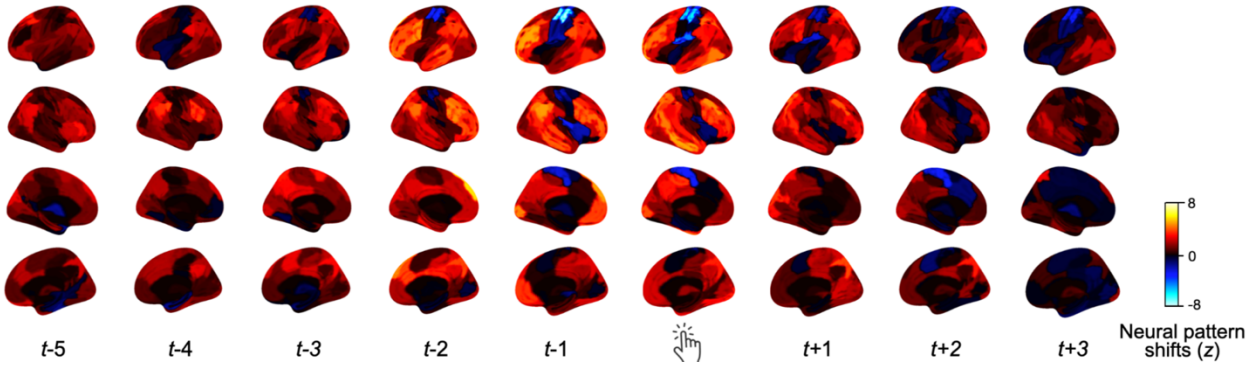


Figure S7. Neural pattern shifts occur near general insight moments. Unthresholded brain maps showing the temporal evolution of neural pattern shifts around general aha button presses. *Z*-statistics indicate deviations from null distributions generated from non-insight moments.

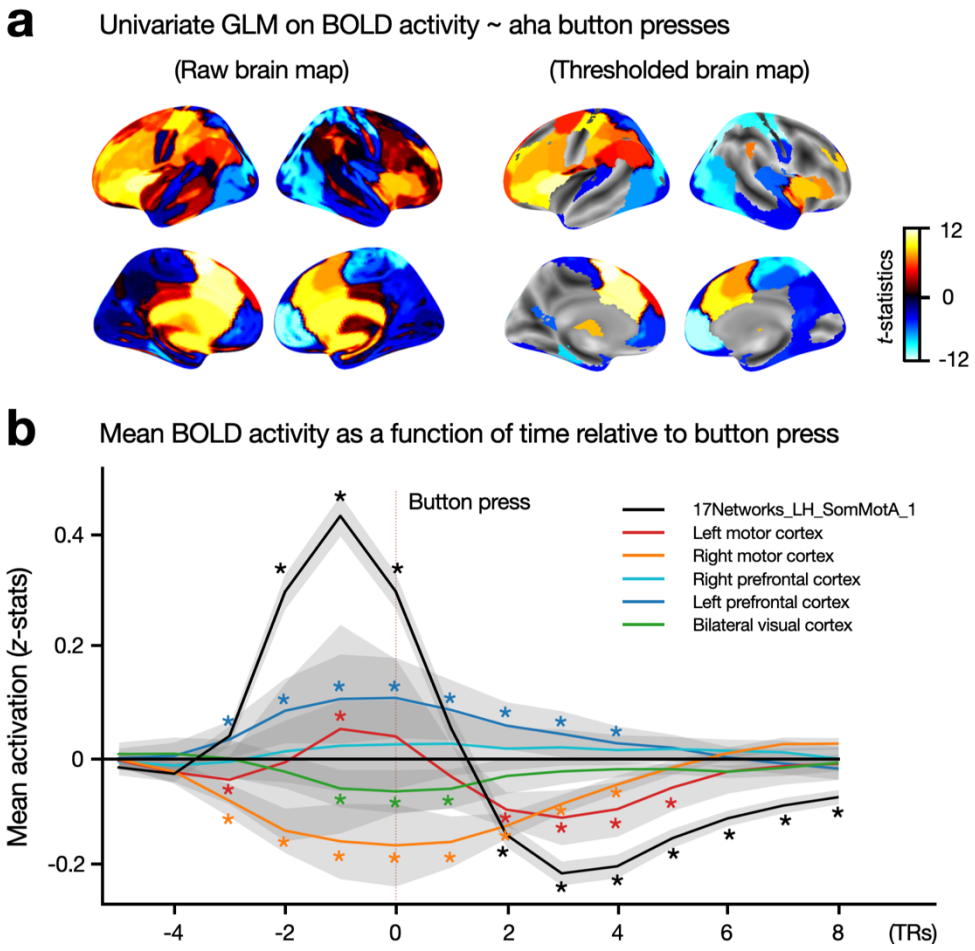


Figure S8. Univariate neural responses to insight moments.(a) Unthresholded (left) and thresholded (right; FDR-corrected $p < 0.001$) brain maps showing ROI-wise *t*-statistics from a univariate GLM analysis, with aha button presses modeled as event onsets shifted by 4 TRs (4.8s) to account for the hemodynamic delay. We observed highly similar results when the aha button presses were modeled as event onsets convolved with the canonical hemodynamic response function and thus did not report them here for brevity. **(b)** Mean BOLD activation relative to aha button presses, shifted by 4 TRs (4.8s) to account for the hemodynamic delay. Shaded areas indicate the SEM across ROIs within each network (e.g., left motor cortex), or across participants for ROI 8. *, $p < 0.05$.

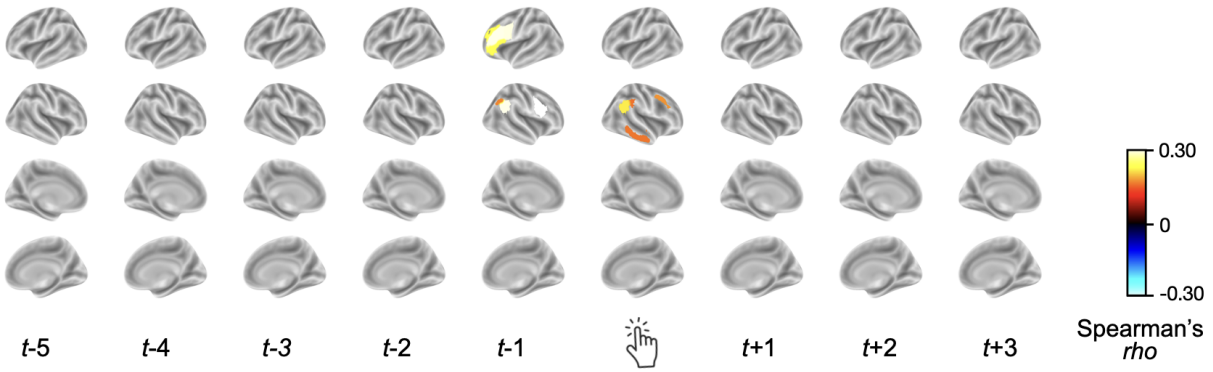


Figure S9. Neural pattern shifts during non-character insights track impression updating. Neural pattern shifts and correlation with impression updating during *non-character aha* moments. ROIs showing both significant neural pattern shifts and their correlation with impression updating. Significance was corrected for multiple comparisons across 116 ROIs and 9 TRs.

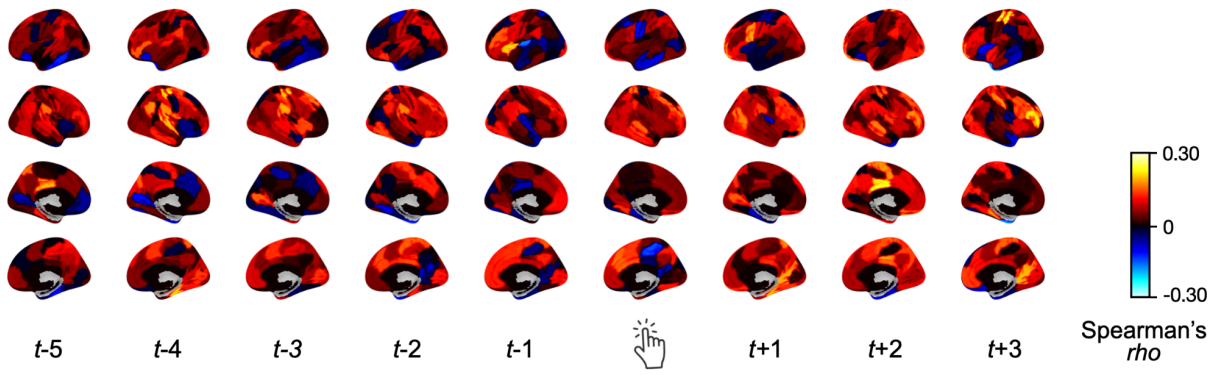


Figure S10. Temporal associations between neural pattern shifts and impression updating during character insights. Raw brain maps of correlations between neural pattern shifts near character insight with impression updating.



Figure S11. Character insight moments are largely distinct from perceptual and semantic scene changes. Venn diagrams showing the overlap between character aha moments and scene changes identified by PySceneDetect (left) and CLIP (middle), as well as the overlap between scene changes identified by the two models (right). Character aha moments showed limited overlap with both PySceneDetect and CLIP scene changes (33.2% for each), whereas the two automated methods showed substantial overlap (79%). These findings suggest that the two methods identify largely similar stimulus-driven changes that are distinct from subjective character insights.

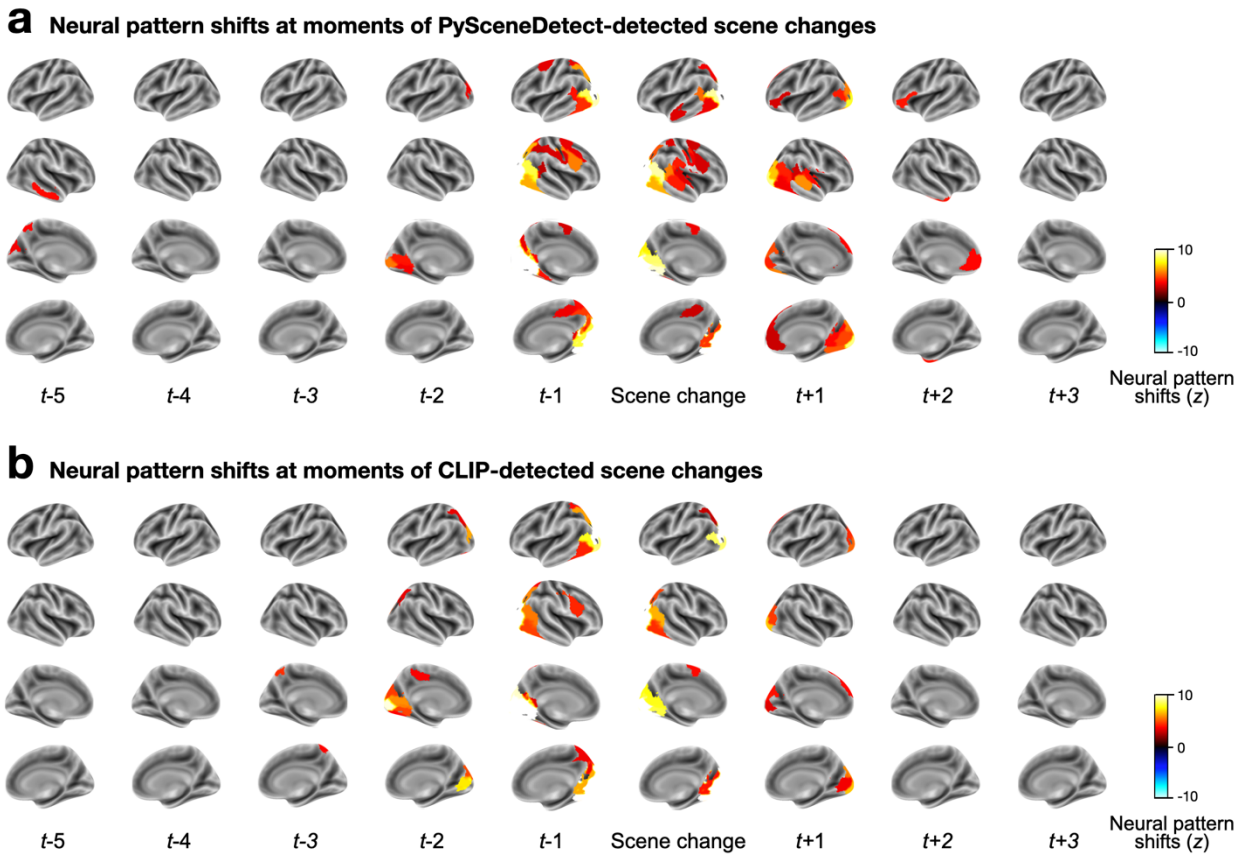


Figure S12. Neural pattern shifts accompany perceptual and semantic scene changes. Neural pattern shifts occurring near moments of scene changes detected by the PySceneDetect **(a)** and CLIP **(b)** models. The brain maps are thresholded maps that show brain regions with significant *z*-statistics, computed by comparing the observed magnitude of pattern shifts with a null distribution derived from non-change moments (FDR-corrected $p < 0.05$).

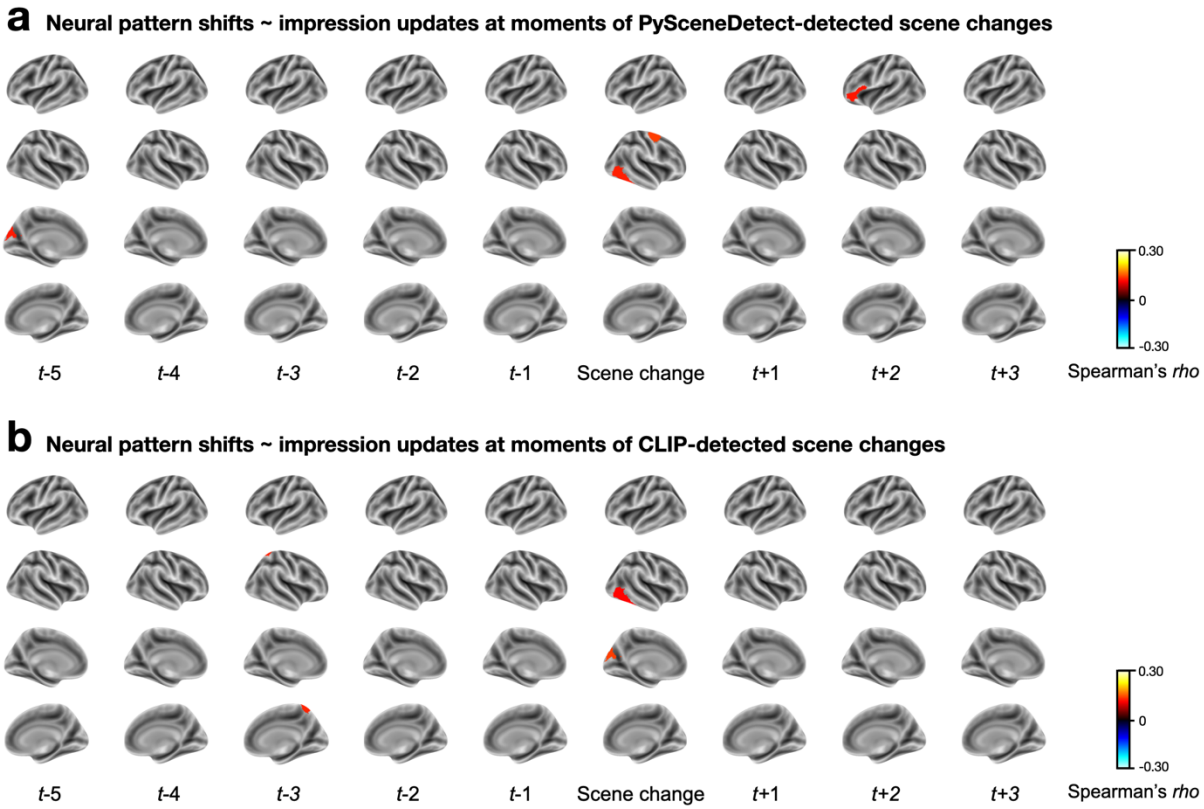


Figure S13. Associations between scene-change-related neural pattern shifts and impression updating. Correlations between impression updating and the neural pattern shifts occur near moments of scene changes detected by the PySceneDetect **(a)** and CLIP **(b)** models. The brain maps show Spearman correlations in regions surviving FDR correction ($p < 0.05$), based on comparisons with a null distribution derived from non-change moments.

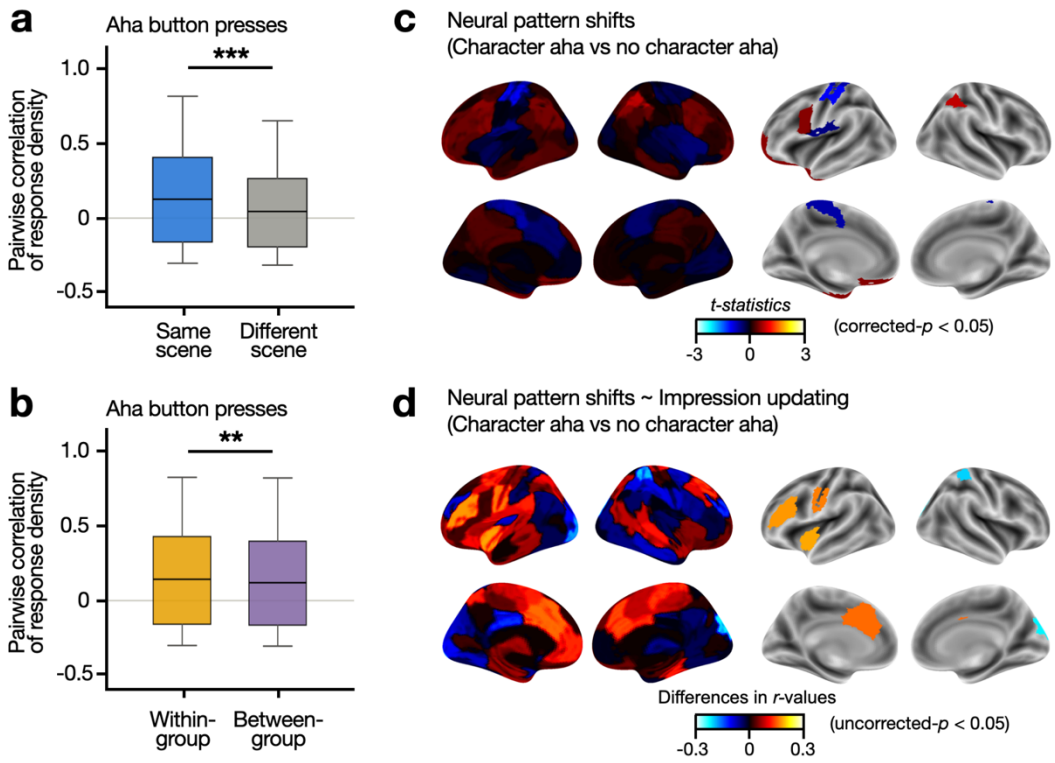


Figure S14. Stimulus-matched analyses distinguish insight-related neural pattern shifts from shared audiovisual responses. (a) Between-participant similarity in aha-button timing was greater within the same scene than across different scenes ($z = 21.7$, $p < 0.001$). Similarity was quantified as the Pearson correlation between participants' Gaussian density profiles of aha-button press times. Boxplots summarize participant-level values; center line, mean; boxes, 25th–75th percentiles; whiskers, 1.5× interquartile range. **(b)** Similarity in aha-button timing between participants viewing the same scene within versus across scrambled-order groups, revealing a smaller but significant effect of narrative context ($t = 2.96$, $p = 0.003$). **(c)** To control for audiovisual input, we compared neural pattern shifts between participants who experienced a character aha moment and those who did not while watching the same scene at the same time point. This allowed us to isolate neural changes associated with subjective insight while holding the movie content constant. Significant aha-related neural pattern shifts were observed in medial frontal and lateral cortical regions. Left, unthresholded *t*-statistic map; right, regions surviving FDR correction ($p < 0.05$). **(d)** Stimulus-matched comparison of associations between neural pattern shifts and impression updating for participants with versus without a character aha moment. Consistent with the original findings, stronger associations were observed in left medial prefrontal and frontotemporal regions, although these effects did not survive multiple-comparison correction. Left, unthresholded map of between-condition differences in *r*; right, regions at uncorrected $p < 0.05$. To increase statistical power while preserving exact stimulus matching, participants were pooled across scrambled-order groups. This stringent analysis required excluding one-third of the scenes and dividing the remaining observations into aha and non-aha conditions, potentially reducing statistical power. n.s., not significant; **, $p < 0.01$; ***, $p < 0.001$.

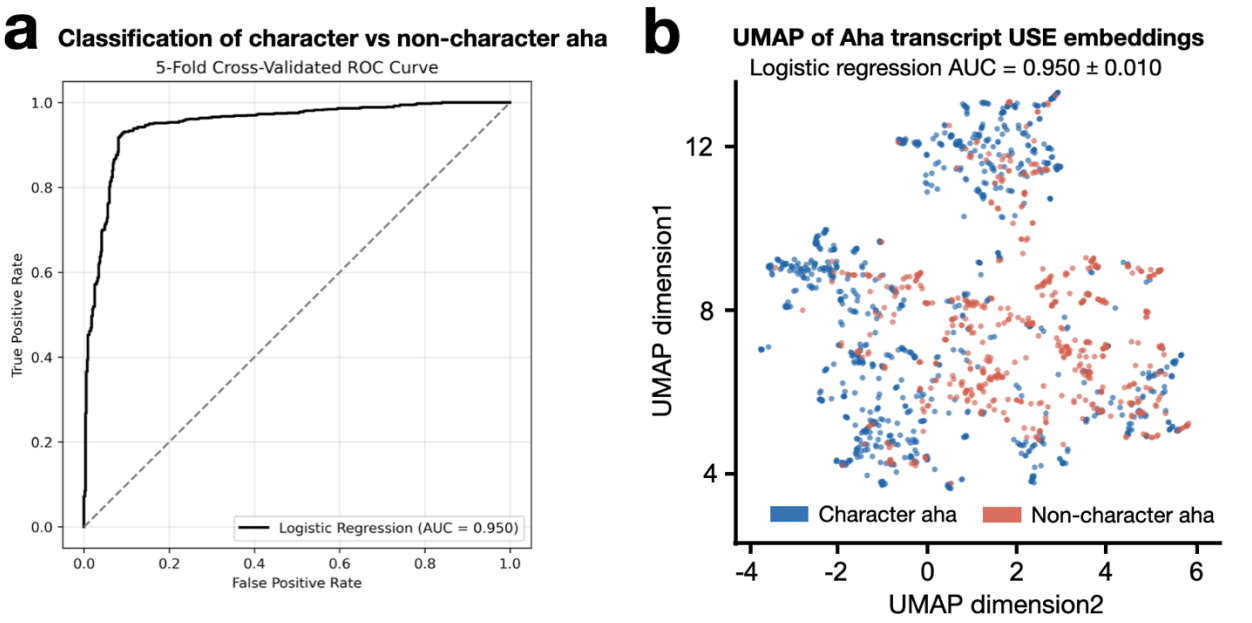


Figure S15. Character and non-character insights exhibit distinct semantic representations. **(a)** Cross-validated classification of USE embeddings of character vs. non-character aha transcripts. Receiver operating characteristic (ROC) curve for the logistic regression classifier evaluated using 5-fold cross-validation. The solid black line represents the average ROC curve across folds, showing the trade-off between true positive rate (sensitivity) and false positive rate (1 − specificity). The dashed diagonal line indicates chance-level performance. **(b)** Universal Sentence Encoder (USE)-embeddings of aha transcripts projected onto the first two dimensions of the Uniform Manifold Approximation and Projection (UMAP) space. Each dot represents an aha verbal report from a single participant, run, and character.

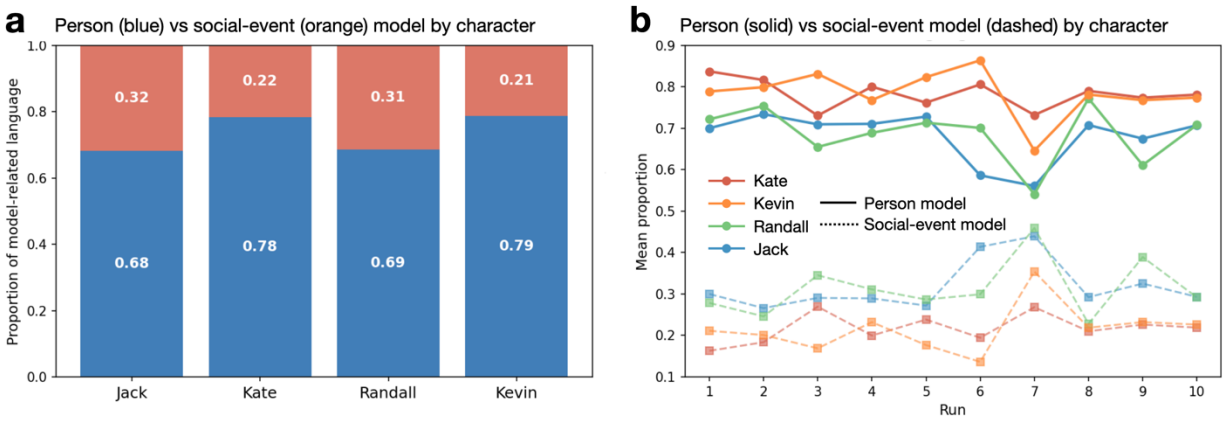


Figure S16. Social impression reports consistently emphasize person-centered representations. Language used for reporting character impressions consistently showed higher person model proportions than social-event model proportions across characters and runs. **(a)** Proportions of person vs social-event model language for each character. Blue indicates person model, orange indicates social-event model. **(b)** Proportions of person vs social-event model language for each character as a function of run. Solid line indicates person model, dashed line indicates social-event model.

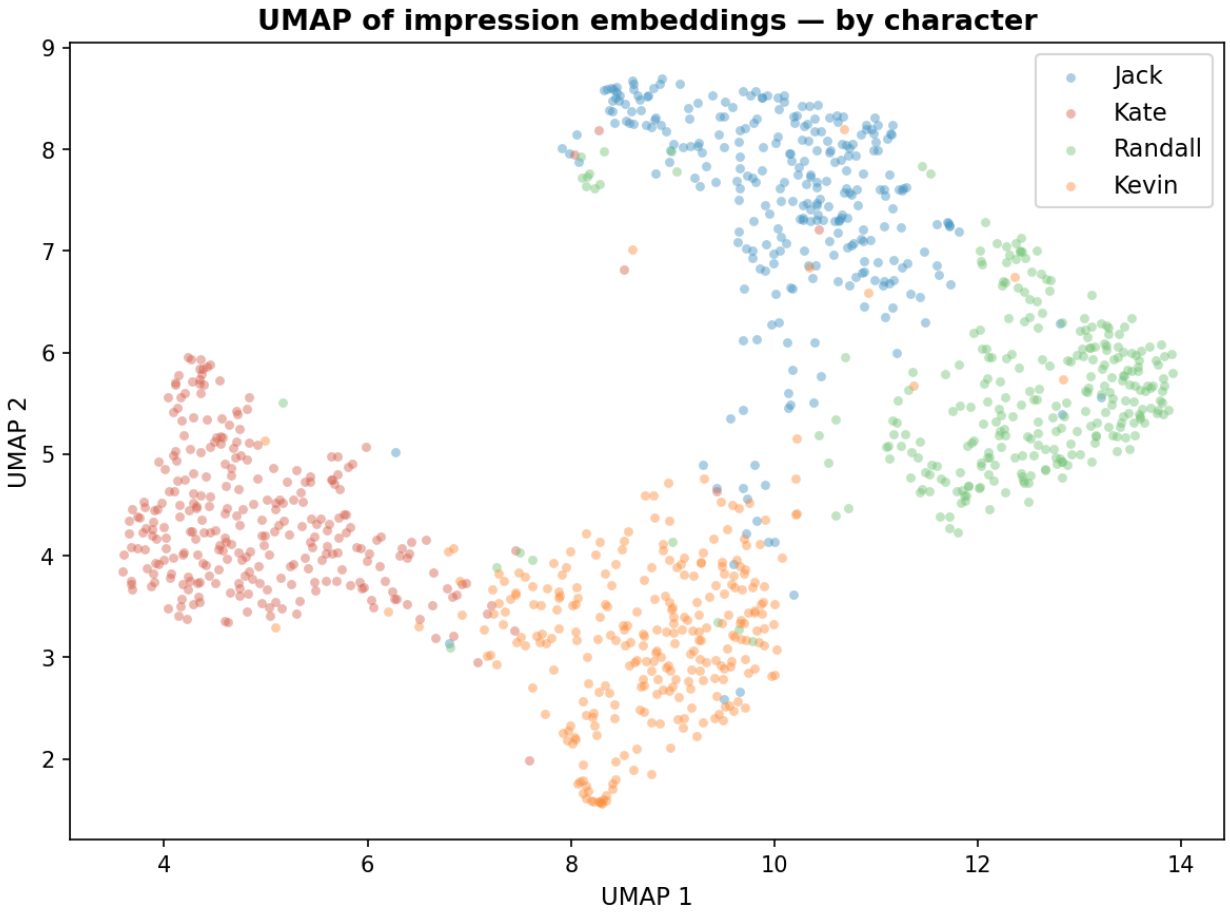


Figure S17. Semantic representations of social impressions differ across characters. Universal sentence encoder (USE) embeddings of impression transcripts projected onto the first two dimensions of the Uniform Manifold Approximation and Projection (UMAP) space. Each dot represents a verbal impression report from a single participant, run, and character.

Supplementary Tables

Table. S1 Definitions of each aha category and their proportions among all aha moments. Verbal explanations for each of the aha button presses from 32 participants were coded into one or more categories (i.e., categories were not mutually exclusive). Categories were derived from participants' responses and applied independently by three raters. For each participant, we calculated the percentage of aha moments assigned to each category (with agreement from >= 2 raters) and then averaged these percentages across participants. In the present study, analyses focused on character-related aha moments. To ensure data quality, we included only those instances that were labeled as "character" by at least two of the three raters (i.e., majority consensus). The non-character aha was defined as the aha moments where none of the three raters labeled as character aha.

Aha category	**Description**	**Mean proportions (SD)** at least 2 raters
Memory retrieval	Explicit or implicit mention of past events	47.2 (19.2) %
Character	Comprehension of the character's traits, behavior, thoughts or feelings	45.8 (19.2) %
Current scene understanding	Comprehension or descriptions of what is happening in the current event	41.2 (17.8) %
Character relationship	Comprehension of the relationship between characters	19.1 (7.6) %
Temporal order	Comprehension of the original temporal order of the events	14.5 (12.6) %
Inference	Inference on narrative contents or connection links between events; Prediction of future events	8.2 (6.4) %
Causal understanding	Comprehension of causal structure of the narrative, or causal relationship between events.	7.3 (5.1) %
Oops	Realizing prior comprehension had been incorrect and/or correcting for it.	4.8 (4.6) %

Table. S2 Segmentation of the full *This Is Us* episode into 48 events, with each event's brief description, featured main character(s), and duration. Segmentation was based on the director's cut, with each scene depicting a single character's narrative except for scenes labeled "Kevin & Kate". These 48 scenes were scrambled in three different orders and presented across 10 fMRI runs, with N=12 per scrambled-order group.

Scene id	Description	Character	Length (s)	Scene id	Description	Character	Length (s)
1	Opening, Birthday towel	Jack	71	25	Leave office	Randall	15
2	Birthday cake in the fridge, Scale	Kate	34	26	Father's house (outside)	Randall	87
3	Work, Good news email, Happy birthday	Randall	26	27	Father's house (inside) 1	Randall	80
4	Bedroom with two girls, I am 36	Kevin	31	28	Father's house (inside) 2	Randall	43
5	Big three, Terrible things to mommy	Jack	30	29	Behind the scene with Alan Thicke	Kevin	33
6	Click good news email	Randall	9	30	A scene with Alan Thicke	Kevin	59
7	Scale naked, Falls back	Kate	18	31	Alternative lighter version	Kevin	54
8	Kiss, My water just broke	Jack	16	32	Meltdown 1	Kevin	54
9	The Challenger story	Kevin	67	33	Meltdown 2: Shame on you, I quit	Kevin	67
10	Bathroom floor, How did we get here?	Kevin & Kate	50	34	Rebecca in labor, Something's not right	Jack	44
11	Bathroom floor, I ate my dream life away	Kevin & Kate	72	35	Father meets family	Randall	51
12	Hospital room, six weeks early birth	Jack	39	36	Cracking up	Randall	38
13	Deep breath, High risk pregnancy	Jack	54	37	Date at a restaurant	Kate	44
14	Dr. Katowski, Best of the best, Bad joke	Jack	66	38	Invite in?	Kate	58
15	Man-ny set, Breastfeed	Kevin	38	39	Pretty picture	Kate	47
16	Ryan Gosling would never do this crap	Kevin	54	40	Toby meets Kevin, meltdown news	Kevin & Kate	55
17	Soccer field with wife and two daughters	Randall	35	41	Doctor shares the news	Jack	86
18	I found him	Randall	48	42	Doctor's story 1	Jack	81
19	Throwing food away, Dog shit	Kate	29	43	Doctor's story 2	Jack	95
20	Group therapy 1	Kate	33	44	I'm dying. Foster parents?	Randall	97
21	Group therapy 2	Kate	58	45	What am I going to do?	Kevin & Kate	72
22	Coffee, You want to be fat friends?	Kate	58	46	Jack and the fireman at the hospital	Jack	61
23	Breathe, I like you two	Jack	40	47	Everyone	Everyone	71
24	Conversation, Three healthy babies	Jack	77	48	The Big Three, Ending	Jack	52

Table. S3 We corrected the mediation effects for multiple comparisons across regions whose synchrony during movie-watching show significant correlations with before and/or after-movie impressions. Here we report the ROI label from the Schaefer 100 atlas, and the statistics from the causal mediation model, including the averaged causal mediation effect (ACME) and its 95% confidence interval, the corresponding *p*-values (uncorrected) and the FDR-corrected *p*-values. **: $p < 0.01$; * $p < 0.05$; † $p < 0.10$.

ROI name	ACME	95% CI	uncorrected-*p*	FDR-*q*
17Networks_LH_VisCent_ExStr_3	0.0009	[-0.0005, 0.0025]	0.214	0.272
17Networks_LH_SomMotB_Aud_1	0.0023	[0.0004, 0.0049]	0.016*	0.072†
17Networks_LH_DorsAttnA_SPL_1	-0.0001	[−0.0012, 0.0011]	0.876	0.876
17Networks_RH_VisCent_ExStr_1	0.0014	[−0.0003, 0.0037]	0.118	0.212
17Networks_RH_SomMotB_S2_1	0.0016	[−0.0001, 0.0037]	0.074†	0.167
17Networks_RH_DorsAttnA_SPL_1	0.0012	[−0.0005, 0.0032]	0.152	0.228
17Networks_RH_DefaultA_IPL_1	0.0013	[−0.0009, 0.0037]	0.242	0.272
17Networks_RH_TempPar_2	0.0030	[0.0008, 0.0060]	0.004**	0.036*
17Networks_RH_TempPar_3	0.0016	[0.0001, 0.0038]	0.028*	0.084†

Table. S4 Word dictionaries used to quantify semantic content in aha and impression transcripts. Words were drawn from prior theory-driven lexicons in social cognition and language research, covering categories such as dispositional traits, mental states, social relationships, causal language, individual-focus pronouns, group-focus pronouns, and event/action terms. Transcripts were tokenized, and the proportion of words matching each category was computed. Categories were combined into two measures: the person model (traits, mental states, individual focus) and the social-event model (relationships, group focus, causal language, events).

Category	Dictionary
Mental state (Tamir et al., 2016)	affection, agitation, alarm, alertness, amazement, ambivalence, amusement, anger, annoyance, anticipation, anxiety, apathy, appreciation, apprehension, attention, awareness, awe, belief, bewilderment, bias, bitterness, boredom, calmness, certainty, cheerfulness, cognition, concern, confusion, consciousness, contemplation, contempt, contentment, curiosity, decision, delight, depression, desire, despair, disappointment, disbelief, disgust, distress, distrust, dominance, dread, dreaminess, drowsiness, earnestness, ecstasy, elation, embarrassment, emotion, empathy, enjoyment, enthusiasm, envy, exasperation, excitement, exhaustion, expectation, fascination, fatigue, feeling, frenzy, friendliness, frustration, fury, gloominess, guilt, hallucination, happiness, horror, humiliation, humor, hunger, hypnosis, hysteria, imagination, impatience, indecisiveness, indifference, insanity, inspiration, intention, interest, intrigue, irritation, jealousy, judgment, laziness, lethargy, loneliness, lust, melancholy, memory, misery, mortification, nervousness, objectivity, opinion, optimism, outrage, pain, panic, patience, peacefulness, pensiveness, pity, planning, playfulness, pleasure, prejudice, preoccupation, pride, rage, reason, regret, relaxation, relief, remorse, resentment, sadness, satisfaction, self-consciousness, self-control, self-pity, serenity, seriousness, shame, shock, skepticism, sleepiness, sorrow, stress, stupor, subordination, surprise, suspicion, sympathy, terror, thirst, tiredness, torpor, trance, transcendence, uncertainty, uneasiness, unhappiness, vengeance, wakefulness, warmth, weariness, woe, worry
Dispositional traits (Thornton and Mitchell, 2018)	warmth, competence, agency, experience, trustworthiness, dominance, openness, conscientiousness, extraversion, agreeableness, neuroticism, attractiveness, intelligence, kind, generous, selfish, ambitious, insecure, confident, shy, honest, dishonest, caring, cold, warm, jealous, loyal, responsible, determined, stubborn, sensitive, emotional, successful, talented, smart, proud, humble, bitter, guilty, ashamed, strong, weak, depressed, anxious, upset, angry, frustrated, passionate, devoted, dedicated, driven, identity, personality, character, nature
Individual focus	he, him, his, she, her, he's, she's
Social relationship	father, dad, mother, mom, son, daughter, brother, sister, sibling, siblings, wife, husband, partner, friend, family, parent, parents, child, children, baby, babies, couple, triplet, triplets, adopted, biological, stepfather, stepmother, marriage, relationship, together, married
Group focus	they, them, their, they're, themselves
Event/Action	happened, occurs, occurred, found, discover, discovered, revealed, reveal, learned, learn, met, meet, told, tell, showed, show, died, dead, born, left, arrived, arrive, went, came, saw, seen, heard, did, done, got, gotten, took, taken, gave, given, brought, sent, lost, won, fell, broke, ran, turned, moved, started, ended, began, finished, decided, chose, chosen, tried, failed, succeeded
Causal language	because, therefore, thus, hence, explains, explanation, reason, caused, resulted, consequently, since, due, why, leads, led, means, implies, suggest, suggests, indicating, indicates